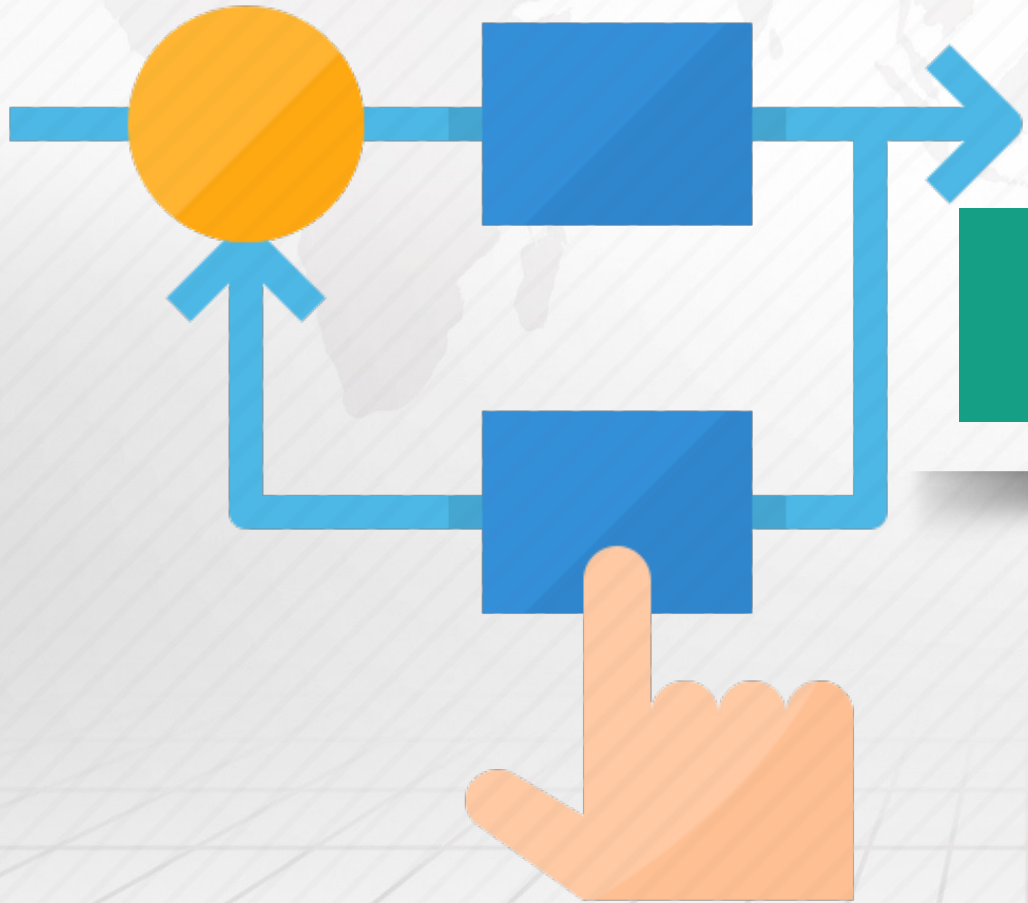




江苏科技大学
JIANGSU UNIVERSITY OF SCIENCE AND TECHNOLOGY



自动控制原理

1. 反馈：控制原理的核心思想

2. 传递函数：控制系统数学模型

3. 方框图：控制系统结构

4. 信号流图：控制系统拓扑结构

5. 梅森公式：数圈圈与消消乐

6. 性能指标：控制效果评价

7. 衰减振荡：欠阻尼二阶系统

8. 稳定：控制系统首要任务

9. 稳态误差：准确性的度量

10. 校正：实现控制的手段

11. 放眼大局：根轨迹法

12. 根轨迹：基本形态

13. 根轨迹：细节修正

14. 参数根轨迹：广义定义

15. 根轨迹法：图形化思考

16. 频率特性：换个角度看控制

17. 近似叠加：绘制伯德图

18. 按图索骥：伯德图求传递函数

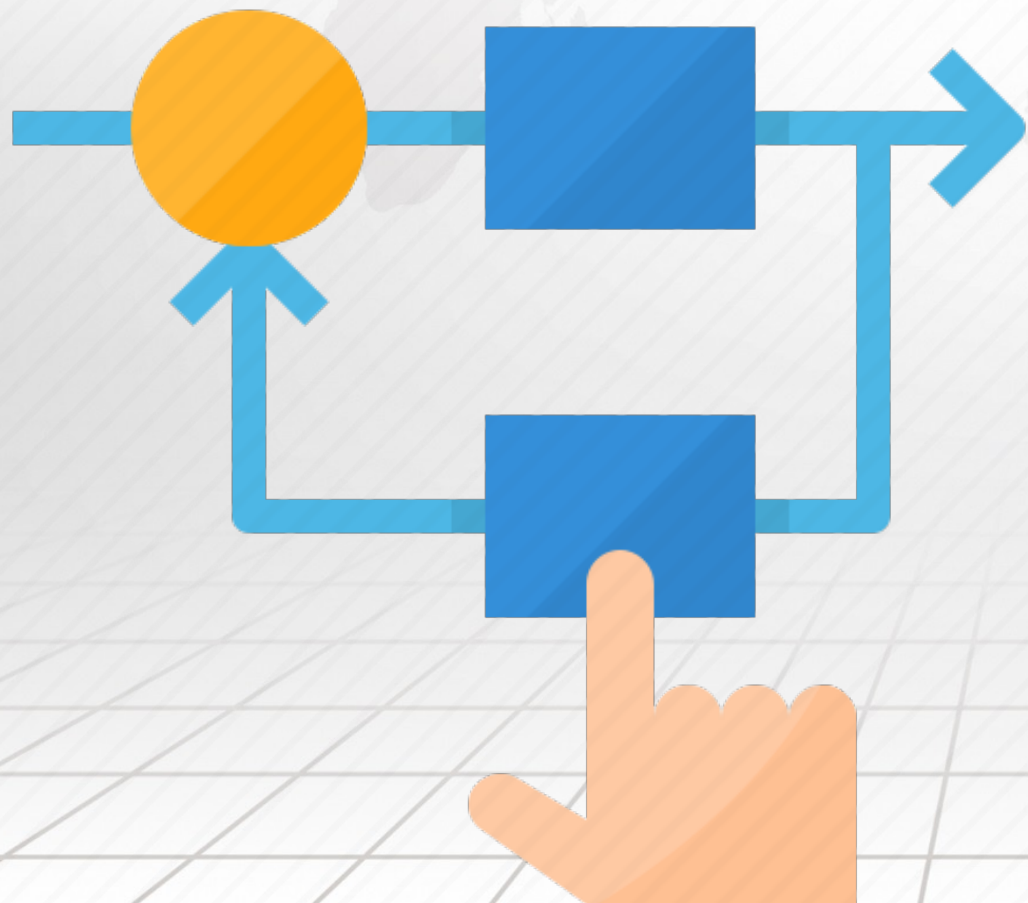
19. 幅相特性：换个角度看频域

20. 宽备窄用：稳定裕度

第一章	第二章	第三章	第四章	第五章	第六章
基本概念	数学模型	时域分析法	根轨迹法	频域分析法	校正方法

5. 梅森公式：数圈圈与消消乐

学习目标



01

能说明梅森公式各部分含义

重

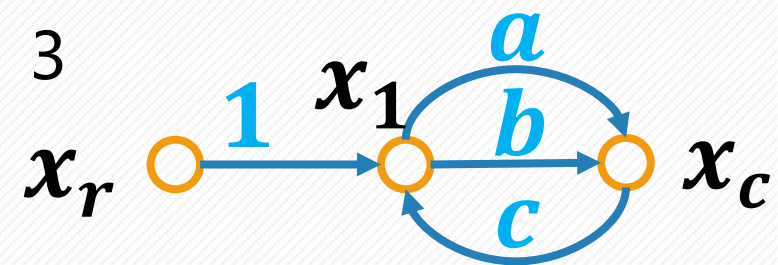
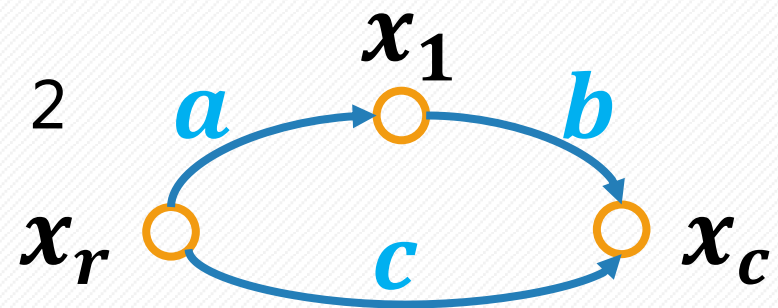
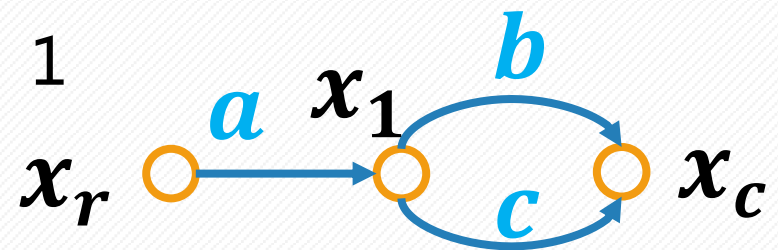
能正确应用梅森公式

03

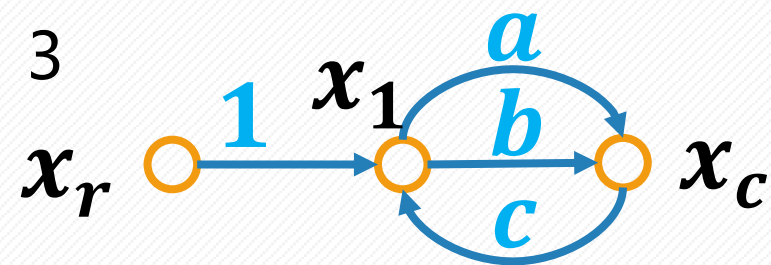
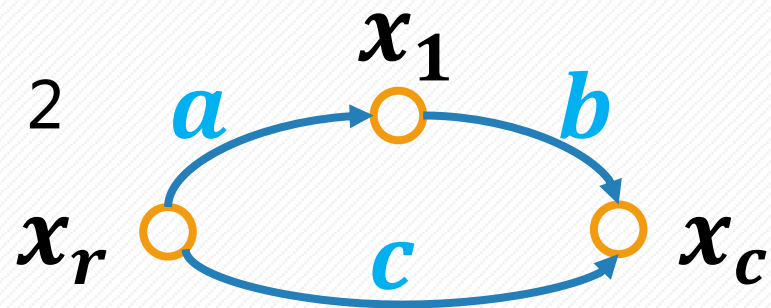
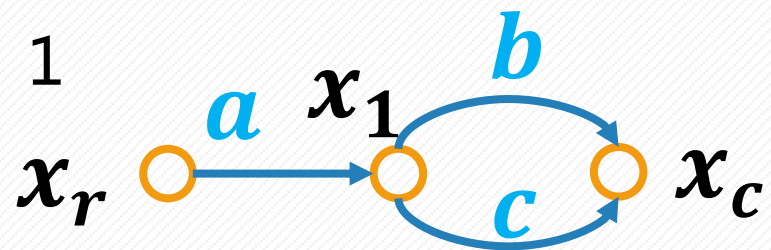
能利用计算机辅助求解传函

难

能求解非零初始条件下响应

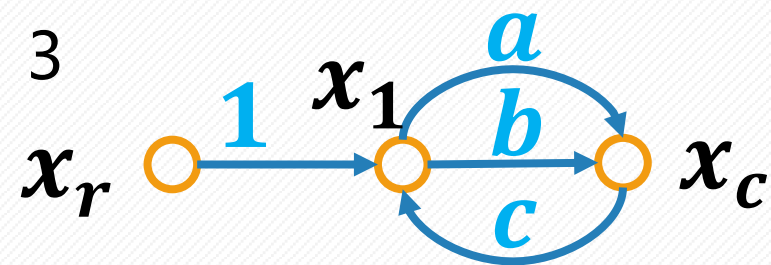
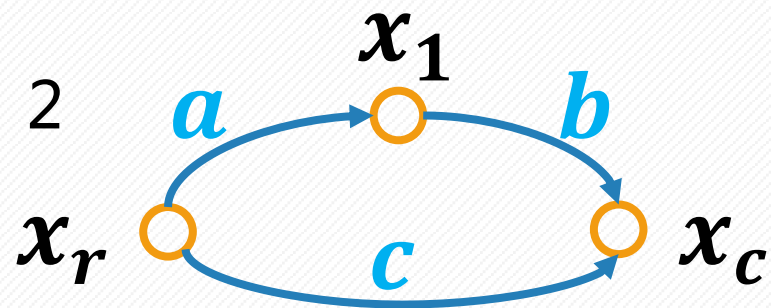
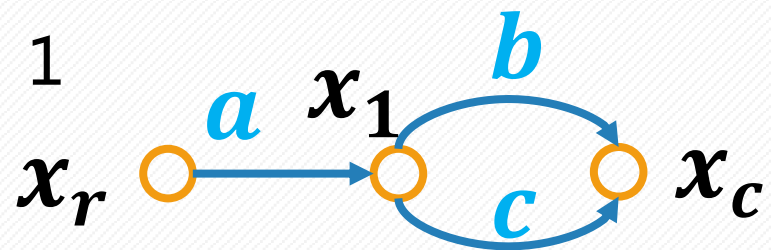


答题区



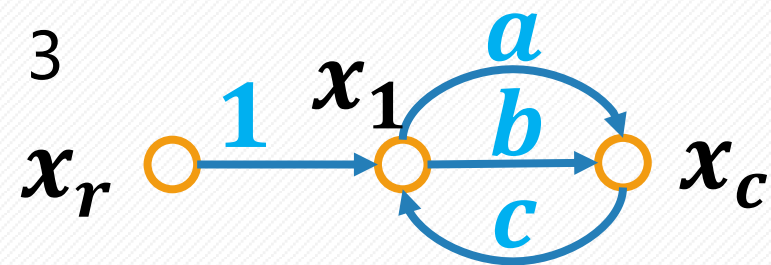
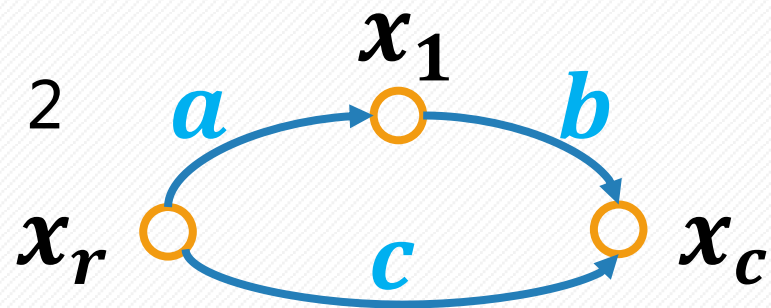
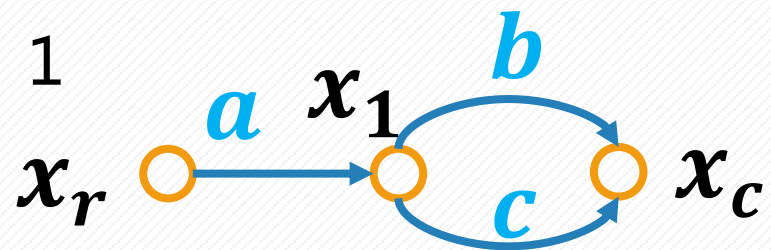
答题区

$$\begin{aligned}
 1 \quad & x_1 = ax_r \quad x_c = bx_1 + cx_1 \\
 & \Rightarrow x_c = abx_r + acx_r = a(b+c)x_r \\
 & \Rightarrow \frac{x_c}{x_r} = a(b+c)
 \end{aligned}$$



答题区

$$\begin{aligned}
 2 \quad & x_1 = ax_r \quad x_c = bx_1 + cx_r \\
 & \Rightarrow x_c = abx_r + cx_r = (ab + c)x_r \\
 & \Rightarrow \frac{x_c}{x_r} = ab + c
 \end{aligned}$$



答题区

$$\begin{aligned}
 3 \quad & x_1 = x_r + cx_c \quad x_c = ax_1 + bx_1 \\
 \Rightarrow & x_c = (a+b)(x_r + cx_c) \\
 \Rightarrow & x_c - (a+b)cx_c = (a+b)x_r \\
 \Rightarrow & \frac{x_c}{x_r} = \frac{a+b}{1-(a+b)c}
 \end{aligned}$$

前向通路：在特征式中除去与某条前向通路相接触的回路后的剩余特征式又被称为前向通路的余子式

前向通路增益

总增益

$$p = \frac{1}{\Delta} \sum_{k=1}^n p_k \Delta_k$$

余子式

特征式

梅森增益公式

$$\Delta = 1 - \sum_a L_a + \sum_{bc} L_b L_c - \sum_{def} L_d L_e L_f + \dots$$

$$\sum_a L_a$$

所有回路增益之和

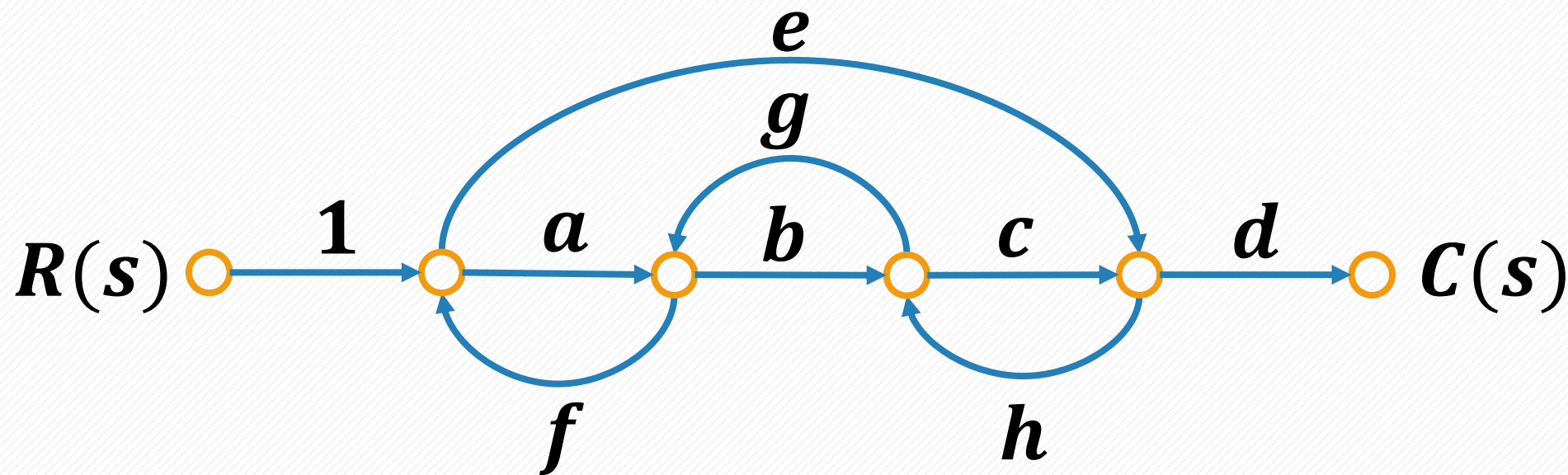
$$\sum_{bc} L_b L_c$$

每两互不接触回路增益乘积之和

$$\sum_{def} L_d L_e L_f$$

每三个互不接触回路增益乘积之和

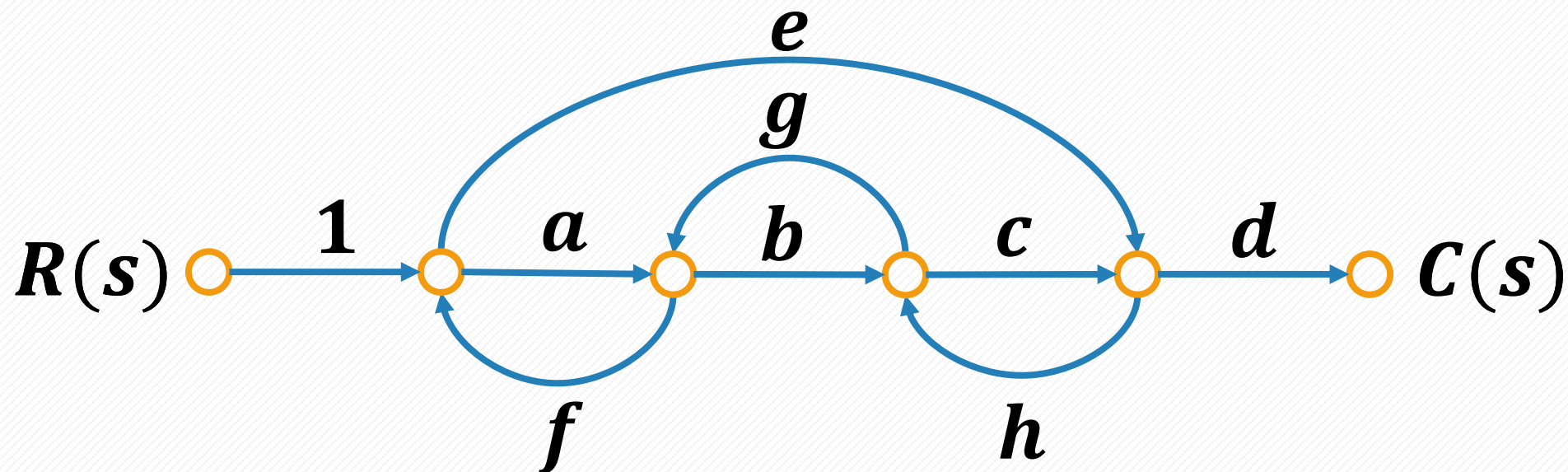
例1：由信号流图求传递函数



回路增益： af bg ch $ehgf$

每两互不接触回路增益： $afch$

例1：由信号流图求传递函数



特征式： $\Delta = 1 - (af + bg + ch + ehgf) + afch$

前向通路增益： $abcd$

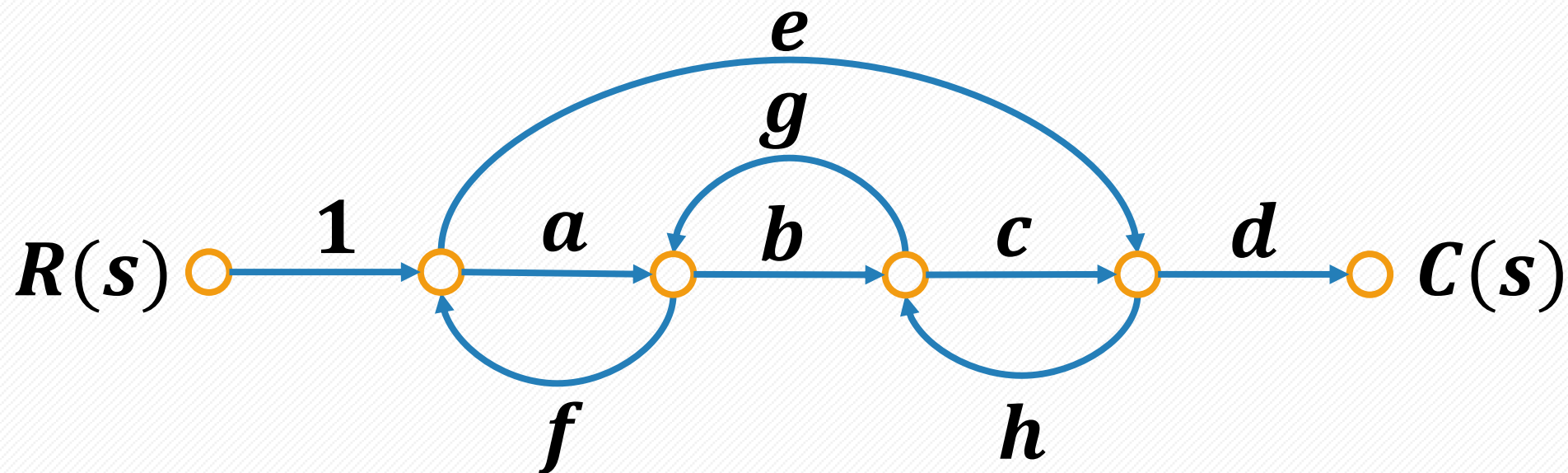
余子式：

1

$1 - bg$

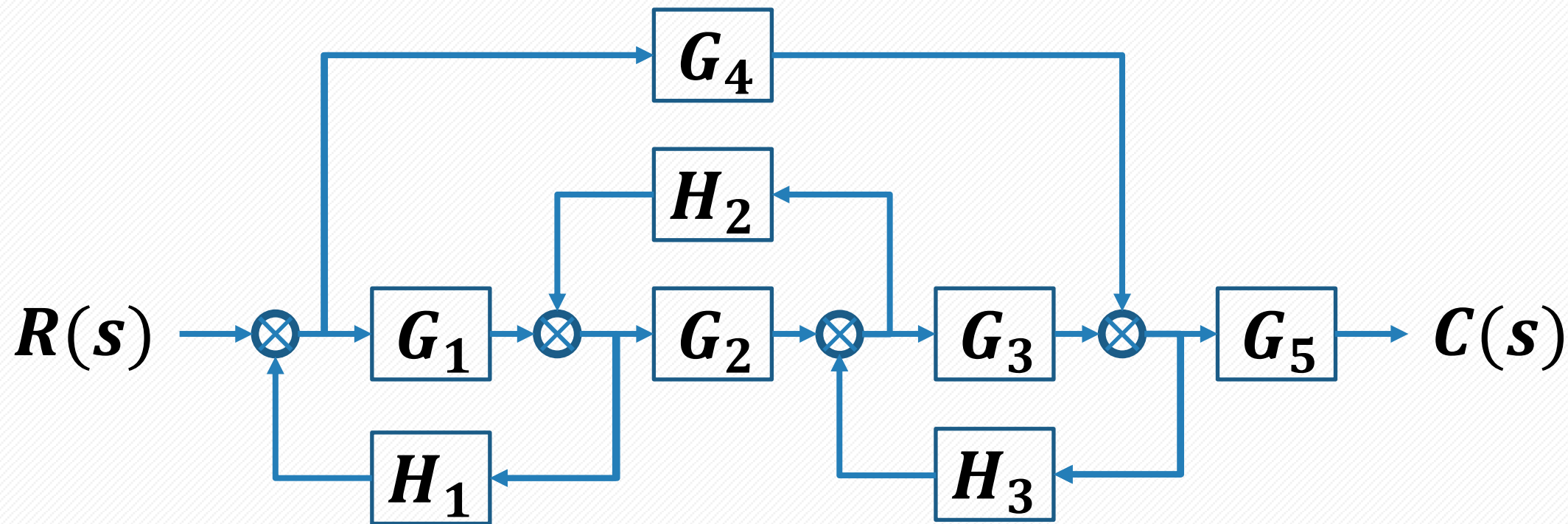
$$\frac{p_1\Delta_1 + p_2\Delta_2}{\Delta}$$

例1：由信号流图求传递函数



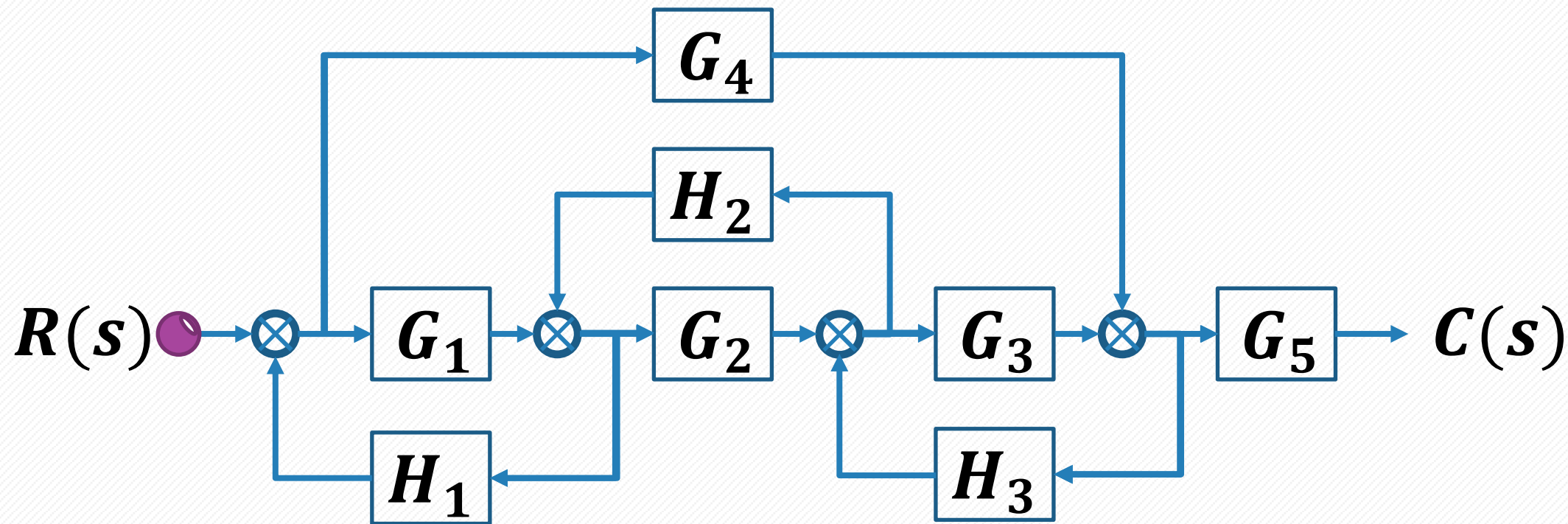
$$\frac{C(s)}{R(s)} = \frac{p_1\Delta_1 + p_2\Delta_2}{\Delta} = \frac{abcd + ed(1 - bg)}{1 - (af + bg + ch + ehgf) + afch}$$

例2：由结构图求传递函数



$$\Delta = 1 - G_1 H_1 - G_2 H_2 - G_3 H_3 - G_4 H_3 H_2 H_1 + G_1 H_1 G_3 H_3$$

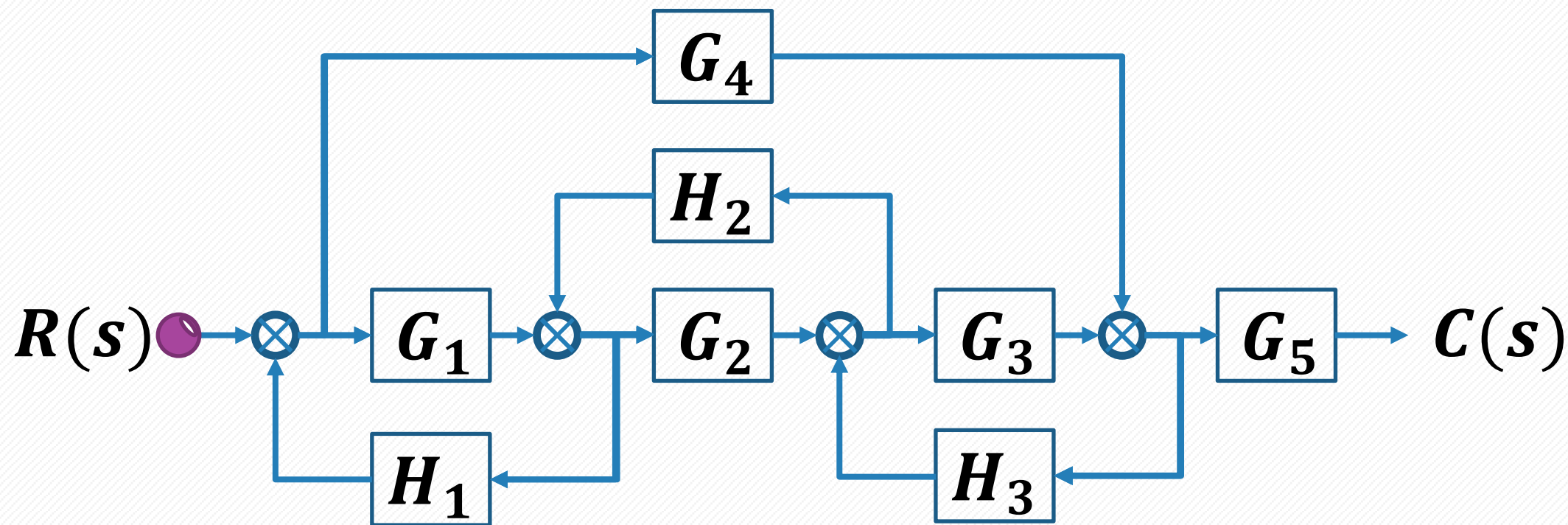
例2：由结构图求传递函数



$$\Delta = 1 - G_1 H_1 - G_2 H_2 - G_3 H_3 - G_4 H_3 H_2 H_1 + G_1 H_1 G_3 H_3$$

$$p_1 = G_1 G_2 G_3 G_5 \quad \Delta_1 = 1$$

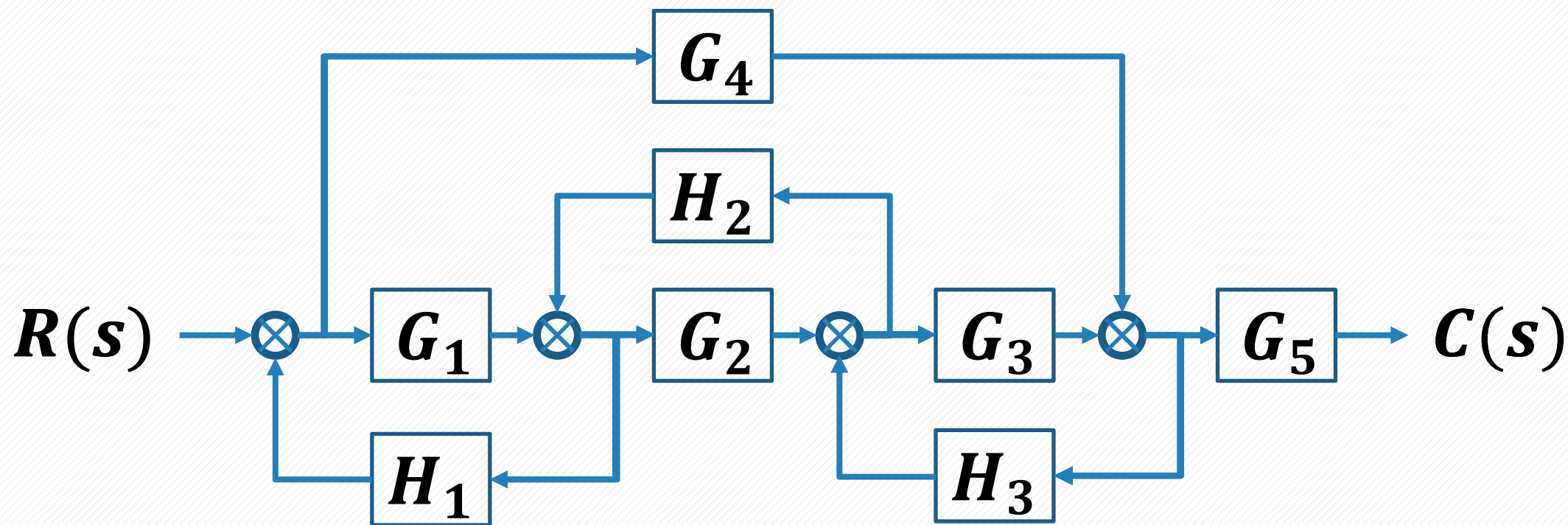
例2：由结构图求传递函数



$$\Delta = 1 - G_1 H_1 - G_2 H_2 - G_3 H_3 - G_4 H_3 H_2 H_1 + G_1 H_1 G_3 H_3$$

$$p_1 = G_1 G_2 G_3 G_5 \quad \Delta_1 = 1 \quad p_2 = G_4 G_5 \quad \Delta_2 = 1 - G_2 H_2$$

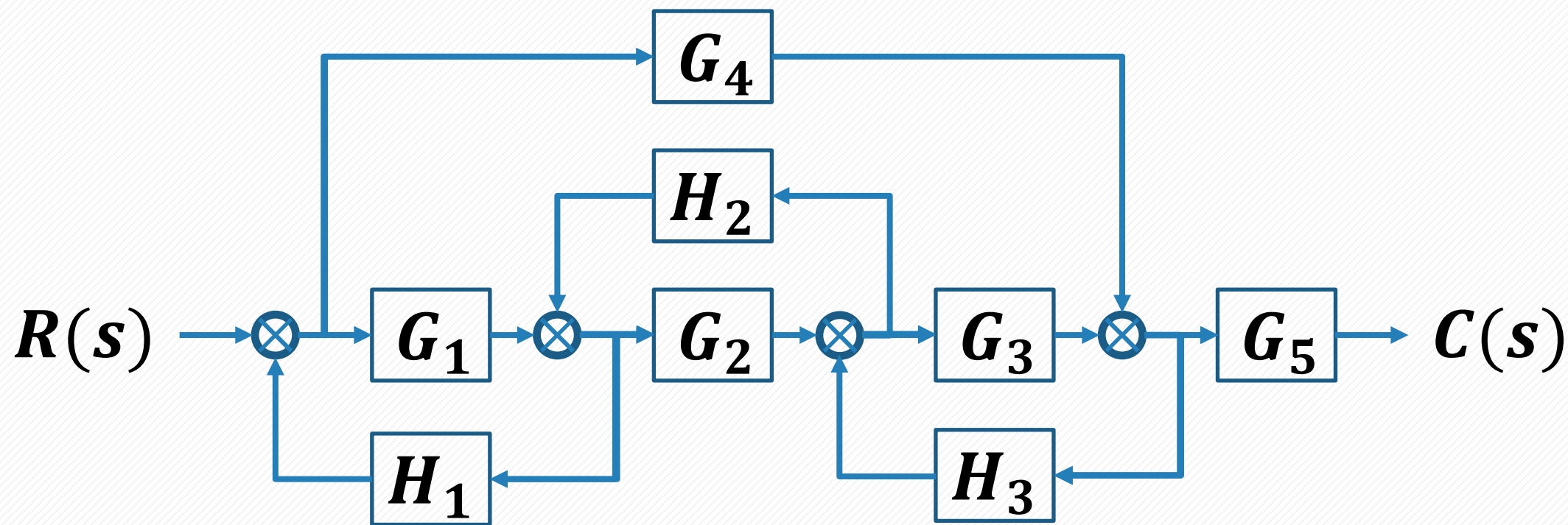
例2：由结构图求传递函数



$$\Delta = 1 - G_1 H_1 - G_2 H_2 - G_3 H_3 - G_4 H_3 H_2 H_1 + G_1 H_1 G_3 H_3$$

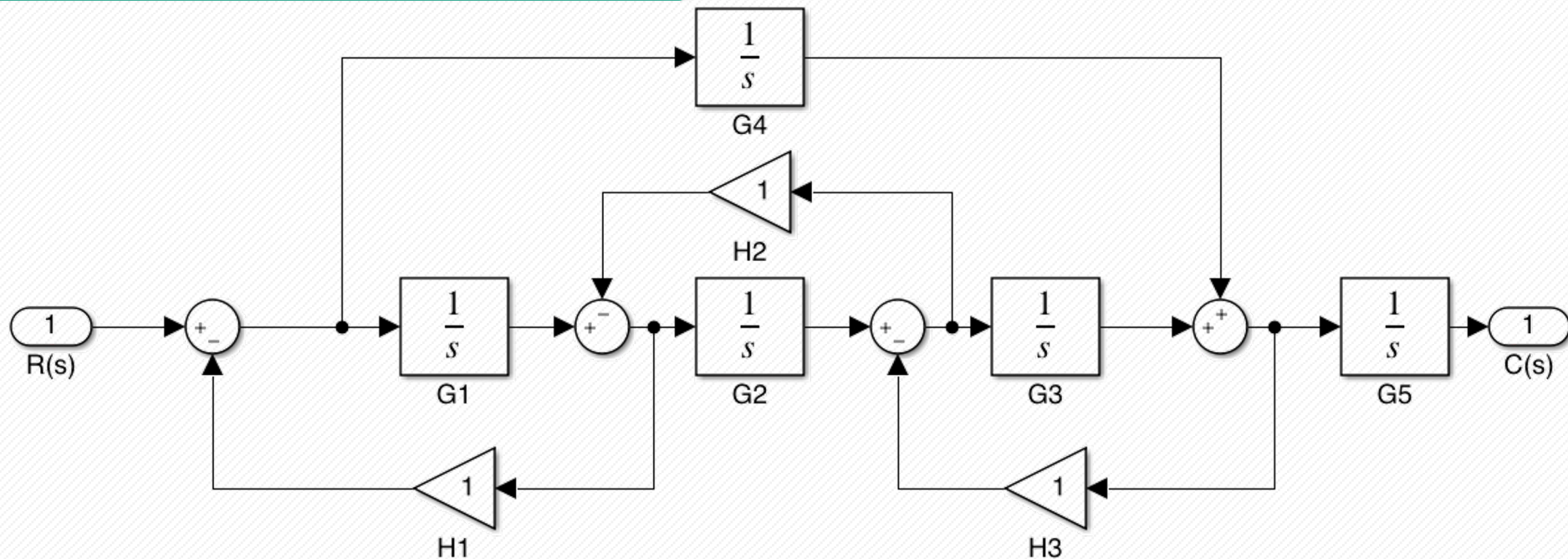
$$p_1 = G_1 G_2 G_3 G_5 \quad \Delta_1 = 1 \quad p_2 = G_4 G_5 \quad \Delta_2 = 1 - G_2 H_2$$

例2：由结构图求传递函数



$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_5 + G_4 G_5 (1 - G_2 H_2)}{1 - G_1 H_1 - G_2 H_2 - G_3 H_3 - G_4 H_3 H_2 H_1 + G_1 H_1 G_3 H_3}$$

例3：计算机辅助求传递函数



$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_5 + G_4 G_5 (1 - G_2 H_2)}{1 - G_1 H_1 - G_2 H_2 - G_3 H_3 - G_4 H_3 H_2 H_1 + G_1 H_1 G_3 H_3}$$

$$\frac{s^3 + s^2 + s - 9.197e-17}{s^5 + 4s^4 + s^3 + 4.25e-16s^2}$$

$$\frac{C(s)}{R(s)} = \frac{s^2 + s + 1}{s^4 + 4s^3 + s^2}$$

拓展：计算机辅助求传递函数

01 基于图论



[1] Shyr-Long Jeng, et. al. A Matrix Approach for Analyzing Signal Flow Graph[J]. Information, 2020, 11, 562

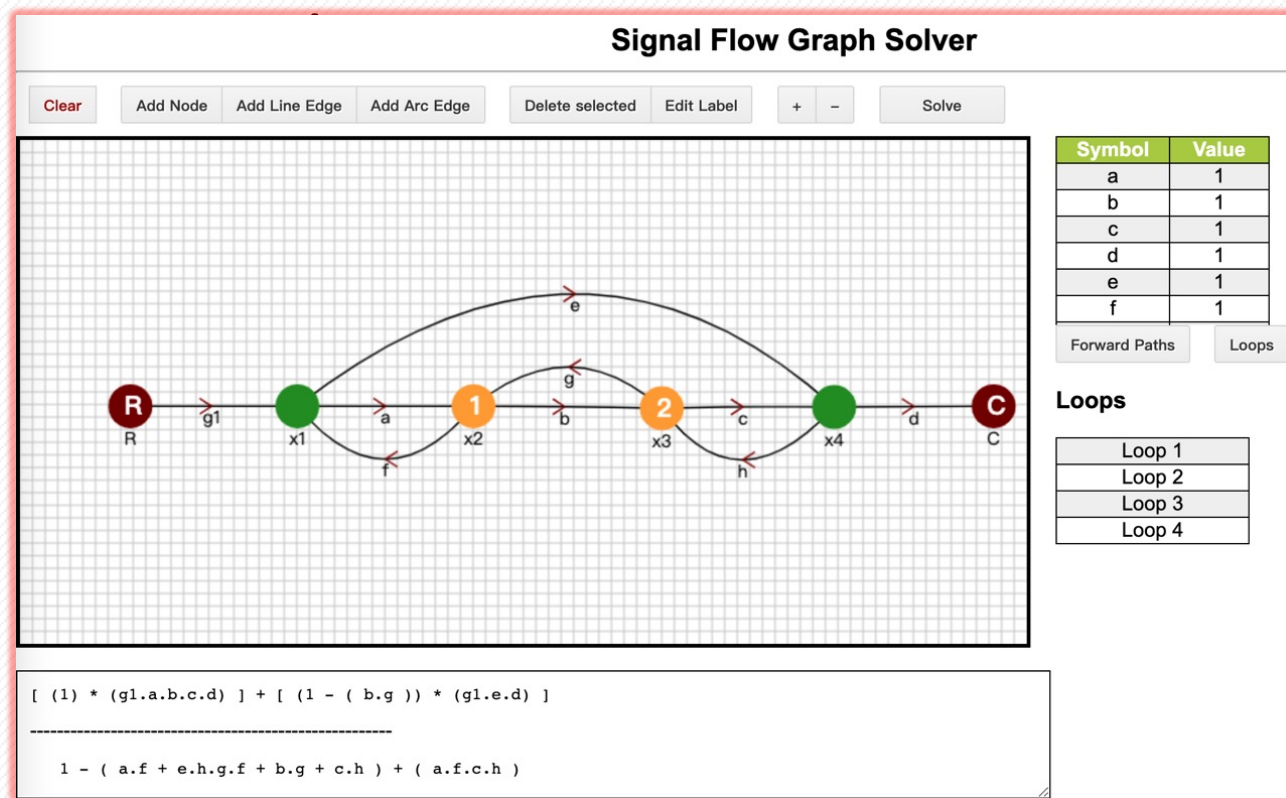
[2] V. C. Prasad. Compact Form of Mason's Gain Formula for Signal Flow Graphs[J]. International Journal of Electrical Engineering Education, 2010, 47/4

拓展：计算机辅助求传递函数

01 基于图论

02 求解器

<https://chrome.google.com/webstore/detail/signal-flow-graph/cdgiabknpabkdlahjgabbamm>



拓展：计算机辅助求传递函数

01 基于图论

02 求解器

03 第三方MATLAB函数



Marko Neitola (2021). A Fully Automated Flowgraph Analysis Tool for MATLAB (<https://www.mathworks.com/matlabcentral/fileexchange/7224-a-fully-automated-flowgraph-analysis-tool-for-matlab>), MATLAB Central File Exchange. Retrieved December 2, 2021.

拓展：计算机辅助求传递函数

01 基于图论

02 求解器

03 第三方MATLAB函数

04 GITHUB项目



<https://github.com/AhmedZahRan7/Signal-Flow-Graph-Solver>

README.md

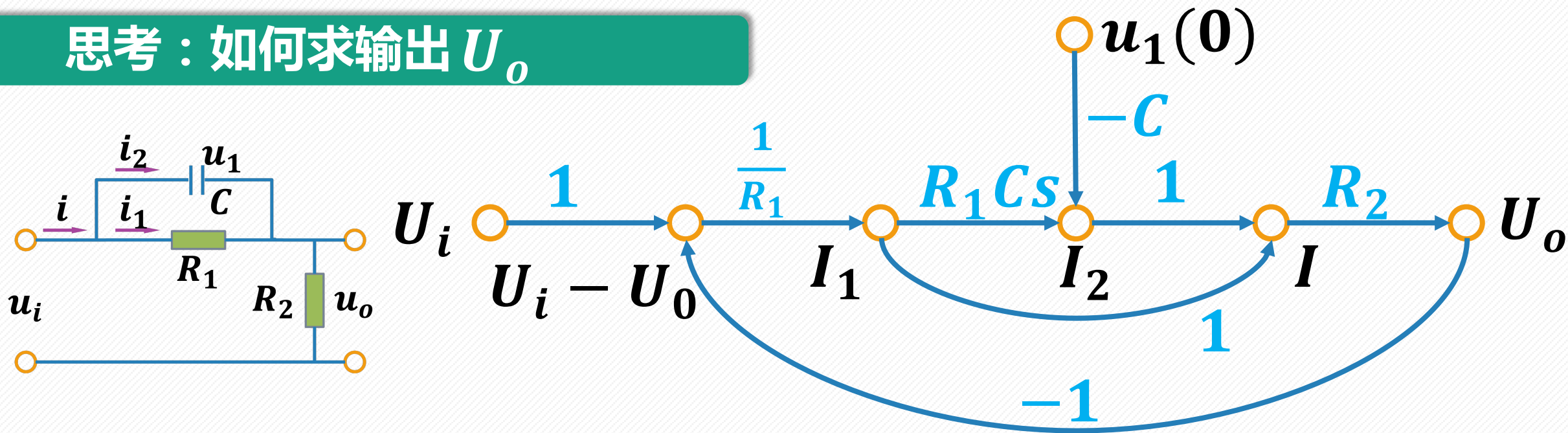
Signal-Flow-Graph-Solver

Signal Flow Graph Solver is a GUI program that represents signal flow graph of different control systems, calculates the overall transfer function and shows it's steps through an interactive user interface.

Abstract

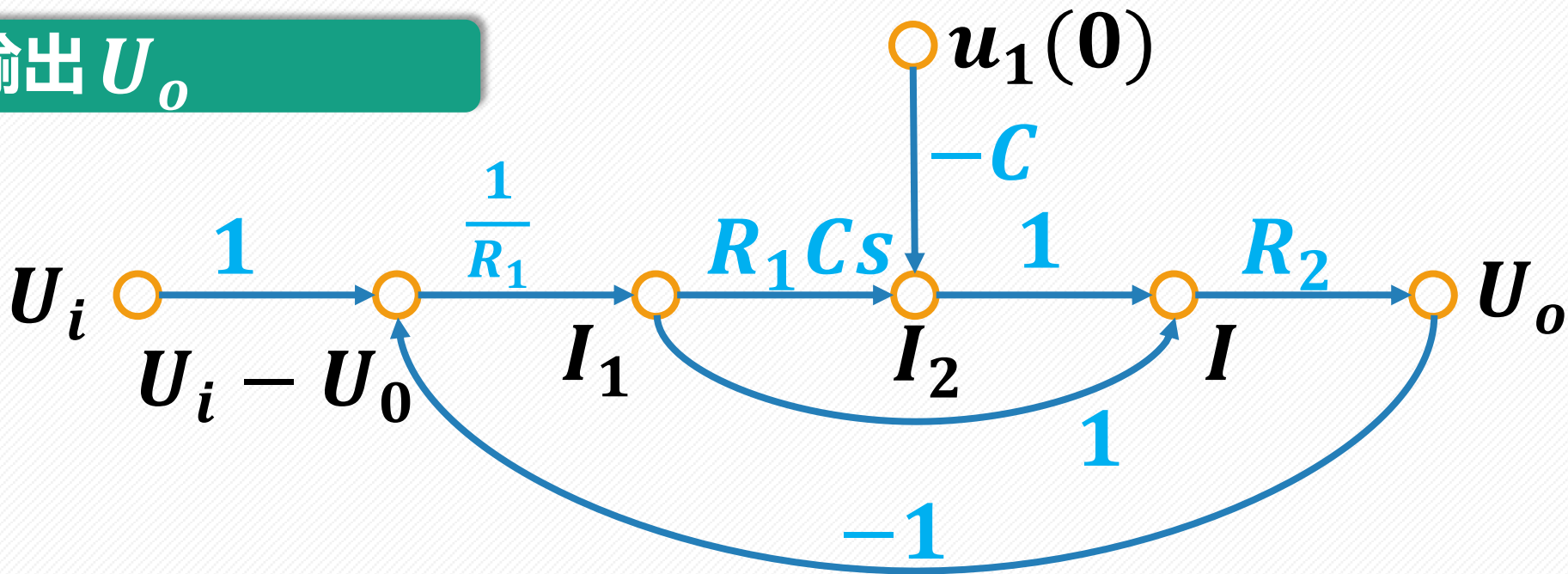
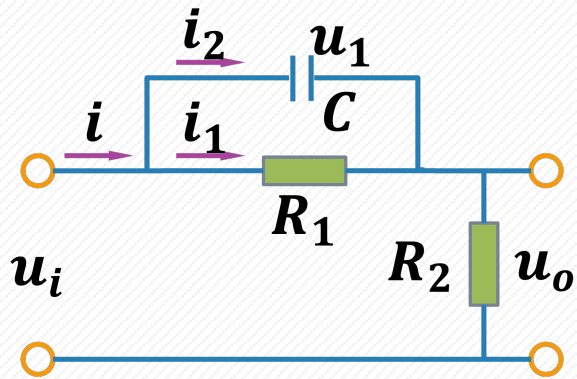
- A signal-flow graph or signal-flowgraph (SFG) is a specialized flow graph, a directed graph in which nodes represent system variables, and branches (edges, arcs, or arrows) represent functional connections between pairs of nodes. Thus, signal-flow graph theory builds on that of directed graphs, which includes as well that of oriented graphs. Mason's gain formula is $T = C(s)/R(s) = \frac{\sum \Delta_i P_i}{\Delta}$ $C(s)$ is the output node $R(s)$ is the input node T is the transfer function or gain between $R(s)$ and $C(s)$ P_i is the i th forward path gain $\Delta = 1 - (\text{sum of all individual loop gains}) + (\text{sum of gain products of all possible two non-touching loops}) - (\text{sum of gain products of all possible three non-touching loops}) + \dots$ Δ_i is obtained from Δ by removing the loops which are touching the i th forward path.

思考：如何求输出 U_o



答题区

思考：如何求输出 U_o



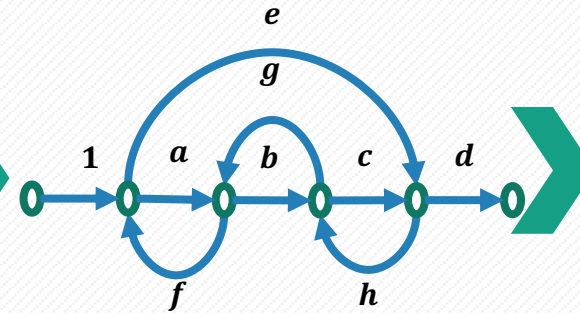
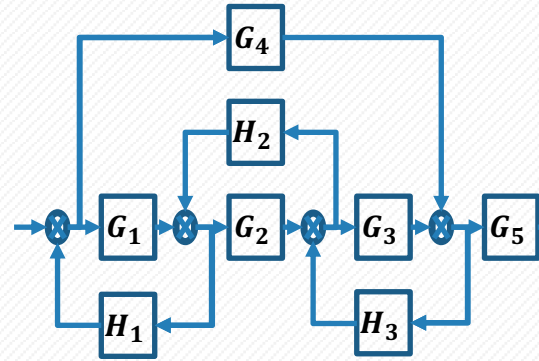
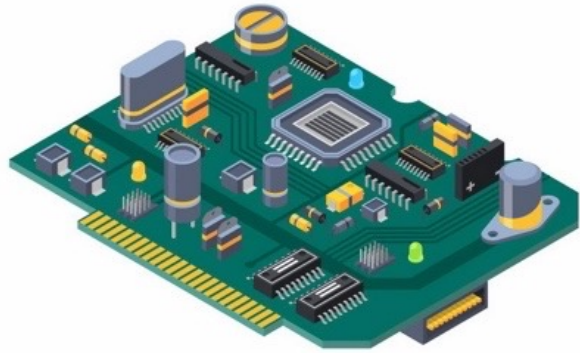
$$\frac{U_o}{U_i} = \frac{CR_2s + \frac{R_2}{R_1}}{CR_2s + \frac{R_2}{R_1} + 1} = \frac{1}{a} \cdot \frac{1 + aTs}{1 + Ts}$$

$$T = \frac{R_1 R_2 C}{R_1 + R_2} \quad a = \frac{R_1 + R_2}{R_2}$$

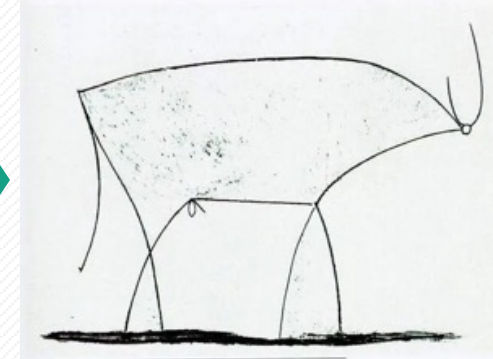
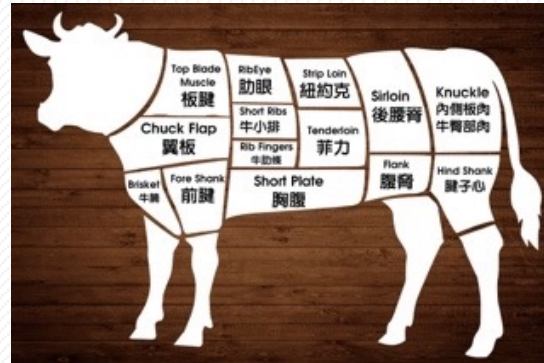
$$\frac{U_o}{u_1(0)} = \frac{-CR_2}{CR_2s + \frac{R_2}{R_1} + 1} = \frac{-T}{1 + Ts}$$

$$U_o = \frac{\frac{1}{a}(1 + aTs)}{1 + Ts} U_i - \frac{T}{1 + Ts} u_1(0)$$

模型抽象



$$\frac{abcd+ed(1-bg)}{1-af-bg-ch-ehgf+afch}$$



Үхэр
 Βοοειδῆ 牛 Eallach
 Vieh Cattle Bovins
 Крупный рогатый скот
 가축 犏

课程小结

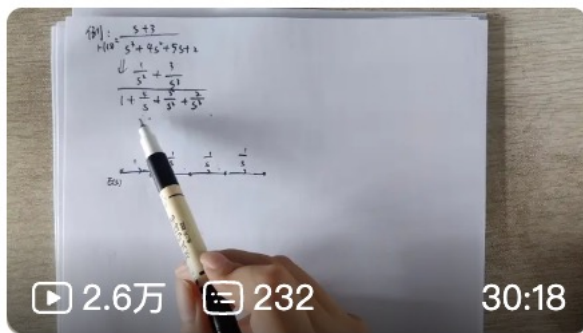
模型	优点	缺点	转传递函数方法	优点	缺点
微分方程	反映机理	计算复杂	拉普拉斯变换	查表可得	复杂情况计算麻烦
结构图	反映结构	画图繁琐	结构图等效	计算简便	部分回路无法解开
信号流图	反映拓扑	高度抽象	梅森增益公式	计算简便	容易漏数回路

$$p = \frac{1}{\Delta} \sum_{k=1}^n p_k \Delta_k$$

“数圈圈”

“消消乐”

课外拓展及预习要求



梅森公式解决信号流图和系统框图问题讲解：
https://www.bilibili.com/video/BV1Qb4y1k7sV?share_source=copy_web

- [1] Mason, S. J. (1956). Feedback theory: Further properties of signal flow graphs.
- [2] Sundararajan, D. (2022). Block Diagrams and Signal-Flow Graphs. In Control Systems (pp. 99-110). Springer, Cham.

5. 梅森公式：数圈圈与消消乐

感谢聆听！



Samuel Jefferson Mason
(1921–1974)